

140 points (70 short questions (14 questions)/ 70 arithmetic tasks (3 tasks))

- 4 Specify requirements (top or further req.)
 - Range
 - Passenger capacity
 - Endurance
 - ETOPS
 - Time to climb
 - Take-off and landing lengths
 - Hangar size
- Explain "Stability Measure" and name a point that is from "immutable" (i.e. aircraft config) and one that can differ from mission to mission

The stability measure characterizes the offset distance of the neutral point to the actual centre of gravity. For a stable aircraft, the stability measure must be negative, that means the AC must be behind the CG.

$$\frac{\partial C_M}{\partial C_L} = \bar{x}_{CG} - \bar{x}_{AC}$$

The CG depends on the mass distribution of the aircraft (structure, payload, fuel, ...)

The AC depends on the aircraft geometry but also on the compressibility effects.

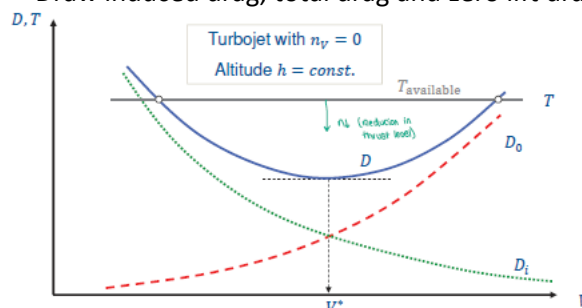
- 3 types of flow separation and which is the most critical regarding stable
 - Trailing edge separation
 - Short bubble separation
 - Long bubble separation

The short bubble separation is the most critical as there is a sudden complete separation.

- Differences engine over / under wing with regard to different aspects (tick which one applies to which config):
 - Cabin Noise → Better under wing
 - Maintainability → Better under wing
 - Risk of damage because things are sucked in (FOD) → Better over wing
 - Noise on Ground → Better over wing
 - Interference during flight → Better under wing
- 2 Identify the difficulties encountering or hampering the introduction of full electric airplanes in the long-range sector
 - Energy density of batteries. Batteries still have a too low energy density meaning that the weight with current batteries would not be viable.
 - Energetic efficiency must be higher (around 75%)

- D/V diagram:

- Draw induced drag, total drag and zero lift drag



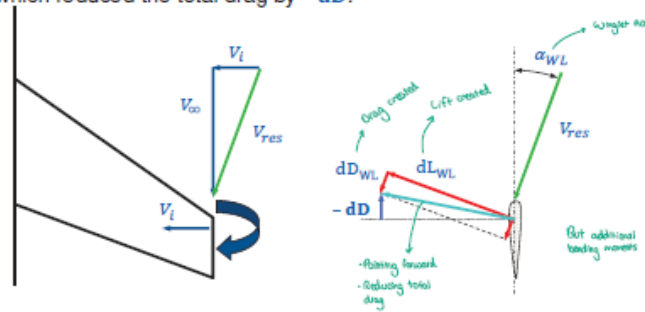
- What to draw with SEP
- Draw optimal L/D ratio (but was believed drag regarding speed)

- Assign terms & explanation: what is net thrust, static thrust, gross thrust, installed thrust, specific impulse, etc.
 - Gross thrust: Impulse force of outflowing mass flow. $T_{gross} = \dot{m}_9 V_9 + (p_9 - p_\infty) A_9$
 - Net thrust: Gross thrust minus the inlet ram drag.

$$T_{net} = \dot{m}_9 V_9 - \dot{m}_2 V_\infty + (p_9 - p_\infty) A_9$$
 - Installed thrust: Net thrust minus the installation losses and additional drag.

$$T_{net,installed} = T_{net} (1 - \Delta P_{bleed}) - \Delta D_{inlet} - \Delta D_{tail}$$
 - Static thrust: Thrust of uninstalled engine at sea level on ISA day.
- What influences MTOW, one point each regarding Atmospheric, Infrastructure Airfield, Plane Configuration before Start
- Design chart given. Record how the 2 curves behave when a profile is used for higher Mach numbers (vertical line for landing and curve for Mach number)
 - The landing limit does not vary with Mach number and the slope of the asymptote of the Mach number limit decreases, allowing for a greater design region.
- Difference between profile or wings for low/high Mach numbers
 - For low Mach aircraft, a laminar airfoil is the most suitable as reduces friction drag. Unswept wings are also desired as there is no compressibility effects.
 - For high Mach aircraft, a supercritical airfoil is the most suitable as they reduce the strength of the shock waves. Swept wings are desired to delay the critical Mach number.
- Given was a 3-page view of a transport aircraft for transonic areas. 4 aspects that suggest that it is a transport aircraft for transonic areas.
 - Swept wings
 - Anti-shock bodies
 - Supercritical airfoil
 - Turbojet/turbofan
- Draw typical flight envelope and enter 2 loads (n_z vs. speed). Point at high speeds enter which with a small angle of attack change the structural (?) Loads. Assume a short, static maneuver
- Indicate when Polhamus, Lifting-Line and Slenderbody are used (on the left the terms, on the right the "explanation" with Ma number and i.what else. Then assign them correctly)
 - Lifting-line theory $\rightarrow AR > 4$ and $M_\infty \ll M_{crit}$
 - Polhamus $\rightarrow AR > 2$ and $M_\infty < M_{crit}$
 - Slender body theory $\rightarrow AR < 2$
- Explain winglet function by means of 2 sketches (once top view on aircraft) and once the winglet (profile)

Winglets use the airflow at the wing side edge, for creating a resulting forward force, which reduced the total drag by $-dD$:



Calculation tasks:

- Aircraft config given, as in exercise (1)
- 1. Task
 - Relation between Maximum Take-Off Weight, Fuel Weight, Payload-Weight und operational empty weight given
 - Diagram given and thus iteratively estimate an MTOW
- 2. Task
 - Wings given with single-slot flap and various diagrams
 - Calculate delta C_l at an angle of attack of x°
 - Calculate C_l at an angle of attack of x°
 - Calculate air density at $h=x$ meters
 - Calculate maximum MTOW
- 3. Task:
 - On and front view (curve flight i.e. with roll angle) given by aircraft
 - Draw thrust, weight, centrifugal, and lift vectors in both views
 - Calculate centrifugal force
 - Calculate Equilibrium point
 - Wings with constant buoyancy distribution given, calculate bending moment